

Basic Terms of Mobile Application Development

1. Android SDK (Software Development Kit)

- **What it is:** A collection of tools, libraries, and resources used to develop Android apps.
 - **Example:** Includes the Android emulator, API libraries, and platform tools.
 - **Tools:** Android Studio, Gradle, SDK Manager.
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2. Android Studio

- **What it is:** The official Integrated Development Environment (IDE) for Android development.
 - **Example:** Used to write, test, and debug Android apps.
 - **Tools:** Inbuilt emulator, code editor, UI designer, and profiler.
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3. Gradle

- **What it is:** A build automation system used to compile, build, and package Android apps.
 - **Example:** Gradle handles dependencies, packaging, and deployment.
 - **Tools:** build.gradle file for project setup.
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4. APK (Android Package Kit)

- **What it is:** The file format used to distribute and install Android apps.
 - **Example:** An APK file is installed on Android devices to run apps.
 - **Tools:** APK Analyzer, Android Studio build tools.
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5. ADB (Android Debug Bridge)

- **What it is:** A command-line tool that allows interaction with an Android device.
 - **Example:** adb install to install APK on a device.
 - **Tools:** ADB commands, Android Device Monitor.
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6. Activities

- **What it is:** A single screen in an Android app where the user interacts.

- **Example:** A login screen, home screen, or settings screen.
 - **Tools:** Activity class in Android.
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7. Intents

- **What it is:** A messaging object used to request an action from another app component.
 - **Example:** Starting a new activity or service.
 - **Tools:** Intent class, startActivity().
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8. Services

- **What it is:** Components that run in the background to perform long-running operations.
 - **Example:** Music playback, network requests.
 - **Tools:** Service class, IntentService.
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9. Broadcast Receivers

- **What it is:** Components that listen for broadcast messages from other apps or system events.
 - **Example:** Listening for network connectivity changes.
 - **Tools:** BroadcastReceiver, IntentFilter.
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10. Content Providers

- **What it is:** Manage access to structured data and share it across apps.
 - **Example:** Accessing contacts, calendar data.
 - **Tools:** ContentResolver, ContentProvider class.
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11. RecyclerView

- **What it is:** A more advanced and flexible version of ListView used to display large datasets.
 - **Example:** A list of chat messages.
 - **Tools:** RecyclerView.Adapter, ViewHolder.
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12. View & ViewGroup

- **What it is:** View is the base class for UI components; ViewGroup is the container for layouts.

- **Example:** TextView, Button, LinearLayout.
 - **Tools:** onCreateView() for layout inflation.
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13. Fragments

- **What it is:** A modular section of an activity that can be reused in different parts of an app.
 - **Example:** A login form fragment or settings fragment.
 - **Tools:** Fragment class, FragmentTransaction.
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14. XML Layouts

- **What it is:** Defines the UI of the app using XML syntax.
 - **Example:** A LinearLayout or RelativeLayout to structure UI elements.
 - **Tools:** activity_main.xml in res/layout.
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15. ViewBinding

- **What it is:** A feature that simplifies interacting with UI elements without findViewById().
 - **Example:** Binding views directly like binding.textView.
 - **Tools:** ViewBinding in Android Studio.
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16. Kotlin

- **What it is:** A modern, concise programming language officially supported for Android development.
 - **Example:** val message = "Hello, World!".
 - **Tools:** Kotlin compiler integrated into Android Studio.
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17. Java

- **What it is:** The traditional programming language for Android development.
 - **Example:** String message = "Hello, World!".
 - **Tools:** Android Studio, JDK.
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18. Jetpack

- **What it is:** A set of Android libraries designed to simplify app development.
 - **Example:** Room for database access, Navigation for app navigation.
 - **Tools:** Room, LiveData, ViewModel.
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19. LiveData

- **What it is:** A lifecycle-aware data holder that can be observed.
 - **Example:** Observing data changes in the UI.
 - **Tools:** LiveData class, Observer.
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20. ViewModel

- **What it is:** A component designed to store and manage UI-related data in a lifecycle-conscious way.
 - **Example:** Storing data between configuration changes.
 - **Tools:** ViewModel class from Jetpack.
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21. Room Database

- **What it is:** An abstraction layer over SQLite, making database access easier.
 - **Example:** Storing user data, settings, etc.
 - **Tools:** @Entity, @Dao, RoomDatabase.
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22. Dependency Injection

- **What it is:** A design pattern used to inject dependencies into classes instead of hardcoding them.
 - **Example:** Using **Dagger** to inject a repository into a ViewModel.
 - **Tools:** **Dagger 2**, **Hilt** (Google's recommended DI library).
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23. Retrofit

- **What it is:** A type-safe HTTP client for Android to handle API requests.
- **Example:** Making network requests like GET, POST.
- **Tools:** Retrofit annotations, Gson for serialization.

24. OkHttp

- **What it is:** An efficient HTTP client used by Retrofit for network communication.
 - **Example:** Handling HTTP requests and responses.
 - **Tools:** OkHttp client, Interceptor.
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25. SharedPreferences

- **What it is:** A simple key-value storage mechanism for saving app settings or user preferences.
 - **Example:** Saving a user's theme preference (light/dark mode).
 - **Tools:** SharedPreferences class, `getSharedPreferences()`.
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26. WorkManager

- **What it is:** A library for managing background tasks that need to be guaranteed to execute.
 - **Example:** Uploading data when the device is connected to Wi-Fi.
 - **Tools:** `WorkRequest`, `OneTimeWorkRequest`.
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27. Permissions

- **What it is:** Requesting and handling permission from users to access certain resources (e.g., camera, location).
 - **Example:** Requesting permission to access the camera using `ActivityCompat.requestPermissions()`.
 - **Tools:** `AndroidManifest.xml`, `PermissionHandler`.
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28. Notification Manager

- **What it is:** A system service that manages notifications.
 - **Example:** Showing a notification when a new message arrives.
 - **Tools:** `NotificationCompat`, `NotificationManager`.
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29. Firebase

- **What it is:** A platform by Google for building and managing mobile apps with various tools like analytics, authentication, and real-time databases.

- **Example:** Using Firebase Authentication for user login.
 - **Tools:** Firebase SDK.
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30. Firebase Firestore

- **What it is:** A NoSQL cloud database used for real-time data syncing.
 - **Example:** Storing chat messages and syncing them in real-time.
 - **Tools:** Firebase SDK, Firestore class.
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31. Firebase Authentication

- **What it is:** A backend service that provides authentication via email, phone, or third-party providers (Google, Facebook).
 - **Example:** Using email/password or Google Sign-In for authentication.
 - **Tools:** FirebaseAuth class.
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32. Google Maps API

- **What it is:** A service that allows developers to integrate maps and geolocation features.
 - **Example:** Displaying maps and adding markers to show user locations.
 - **Tools:** GoogleMap, MarkerOptions.
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33. ConstraintLayout

- **What it is:** A flexible layout that allows you to create complex UI with flat view hierarchies.
 - **Example:** Positioning views relative to each other using constraints.
 - **Tools:** ConstraintLayout widget.
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34. Android Emulator

- **What it is:** A tool to test Android apps on different virtual devices.
- **Example:** Running an app on an emulator

with different screen sizes and Android versions.

- **Tools:** Android Virtual Device (AVD) Manager.
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35. UI/UX Design

- **What it is:** Designing the user interface and experience for ease of use and visual appeal.
 - **Example:** Creating wireframes, interactive prototypes.
 - **Tools:** Figma, Sketch, Adobe XD.
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36. Multi-threading

- **What it is:** Running multiple operations in parallel to improve app performance.
 - **Example:** Fetching data from a server on a background thread while keeping the UI responsive.
 - **Tools:** AsyncTask, ExecutorService, Coroutine (Kotlin).
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37. Kotlin Coroutines

- **What it is:** A Kotlin feature for writing asynchronous code in a more readable manner.
 - **Example:** Launching a background task to download data.
 - **Tools:** GlobalScope.launch, async.
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38. UI Testing

- **What it is:** Automating the testing of the user interface to check app behavior.
 - **Example:** Verifying if a button click takes the user to the correct screen.
 - **Tools:** Espresso, UI Automator.
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39. Unit Testing

- **What it is:** Testing individual components or functions in isolation.
 - **Example:** Testing if a function that calculates the sum works correctly.
 - **Tools:** JUnit, Mockito.
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40. Proguard

- **What it is:** A tool for code shrinking, obfuscation, and optimization to make the APK smaller and protect code.
- **Example:** Minimizing the size of an app's APK.
- **Tools:** proguard-rules.pro.

41. Crashlytics

- **What it is:** A Firebase tool for tracking crashes and issues in real-time.
 - **Example:** Monitoring and fixing app crashes by getting detailed reports.
 - **Tools:** Firebase Crashlytics SDK.
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42. Lambda Functions

- **What it is:** Anonymous functions in Kotlin, useful for handling simple operations.
 - **Example:** `val sum = { a: Int, b: Int -> a + b }`.
 - **Tools:** Kotlin standard library.
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43. Dagger 2

- **What it is:** A dependency injection framework that helps manage dependencies in large apps.
 - **Example:** Injecting repositories and services into ViewModels.
 - **Tools:** `@Inject`, `@Component`.
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44. Hilt

- **What it is:** A newer, simplified DI library built on top of Dagger 2, specifically for Android.
 - **Example:** Simplifying DI with minimal setup.
 - **Tools:** `@HiltAndroidApp`, `@Inject`.
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45. Material Design

- **What it is:** A design language for Android that ensures a consistent and visually appealing UI across apps.
 - **Example:** Implementing Material Buttons, Cards, and Bottom Navigation.
 - **Tools:** Material Components library.
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46. Splash Screen

- **What it is:** The introductory screen that appears when an app is launched.
- **Example:** A logo or animated image shown briefly during app startup.

- **Tools:** SplashScreen API in Android 12+.
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47. AdMob

- **What it is:** A mobile advertising platform by Google to monetize apps with ads.
 - **Example:** Displaying banner ads or interstitial ads in an app.
 - **Tools:** AdMob SDK.
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48. Analytics

- **What it is:** Collecting and analyzing data to understand user behavior and improve app performance.
 - **Example:** Tracking user events like button clicks or screen views.
 - **Tools:** Firebase Analytics, Google Analytics.
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49. App Signing

- **What it is:** The process of digitally signing an APK before it can be published.
 - **Example:** Using a keystore file to sign an APK.
 - **Tools:** keytool, Android Studio signing options.
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50. Google Play Console

- **What it is:** The platform for managing Android app distribution, updates, and monitoring.
- **Example:** Uploading APKs, managing reviews, and tracking app performance.
- **Tools:** Google Play Console dashboard.